DoangJoo "Alan" Synn

Ph.D. Student in Computer Science · Georgia Institute of Technology
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Research Interests

My research explores computational design and creativity, with a focus on representing and supporting creative intent in generative and interactive systems. I build models and interfaces that translate high-level human intent into expressive, controllable outcomes, spanning domains such as computer graphics, motion synthesis, and kinematic design.

Education

Georgia Tech, Ph.D. in Computer Science — Aug. 2022-Present
- School of Interactive Computing; advised by Profs. Oh & Ha
- Research: computational design, mechanical artifacts, and motion synthesis (HCI/Graphics)

Korea University, M.Eng. in ECE — Mar. 2020-Feb. 2022
- Thesis: *On Efficient Data Delivery for Machine Learning Systems*

Korea University, B.Eng. in EE — Mar. 2012-Feb. 2020

Experience

Ph.D. Research Intern, Dolby Laboratories — May 2024-Aug. 2025
- Built a time-series, multimodal datastore supporting low-latency media analytics in the Experience Delivery Lab.

MLOps Engineering Intern, LG Uplus Corp. — June-Aug. 2023
- Led **DeepCrunch**, a model-compression library supporting multi-backend AI inference for efficient edge deployment.

Research Scientist, Protopia AI Inc. — Jan. 2021-Aug. 2022
- Built **Stained-Glass**, a privacy-preserving deep-learning framework enabling inference without raw-data exposure.

Cloud Architect · AWS Educate Cloud Ambassador — 2020-2022
- Led engineering for four startup services, translating ambiguous product requirements into production backend systems.

Server Engineer, Flysher Inc. — Jan.-Dec. 2019
- Shipped backend features for four social-game titles; one reached Facebook's global top 10.

System Engineer, AKA Intelligence (Musio) — Apr. 2017-Jan. 2019
- Designed embedded Linux boards and firmware for **Musio**, a deployed social robot.

Selected Publications

1 **DoangJoo Synn**, Zhifan Guo, Sehoon Ha, and HyunJoo Oh. "MotionSmith: A Sketch-Based Design System for Automata Making." *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, 2026. doi:10.1145/3772318.3791545.

2 Jin Yu, Poojita Garg, **DoangJoo Synn**, and HyunJoo Oh. "Tangible-MakeCode: Bridging Physical Coding Blocks with a Web-Based Programming Interface for Collaborative and Extensible Learning." *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, 2025. doi:10.1145/3706598.3713260.